

D.C. Electric Circuits

current: transfer of positive charge
(I) per unit time

$$I = \frac{dQ}{dt}$$

- .. always defined through devices
(analogous to fluid flow)
- .. has magnitude in amperes
and direction, labelled w/ arrow

Voltage:
(V) .. necessary for current flow
a/k/a potential
.. analogous to pressure
.. always defined across devices
i.e., difference or drop
.. has magnitude in volts
and polarity, indicated w/
+ and - signs

Power : for now: $P = VI$

(P)

.. magnitude in watts (W)

.. sign tells us whether power is being created or absorbed

Sign convention

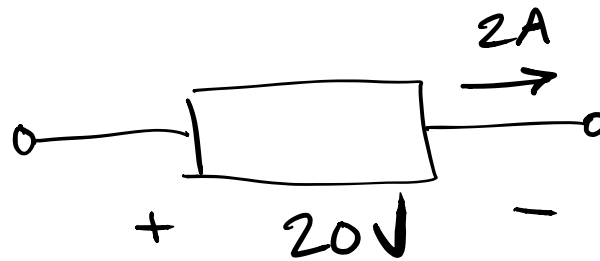
1.) voltage drops are positive w/
current flow

(voltage rises are negative)

2.) Positive power $\rightarrow$ dissipation
(absorbed)

Negative power $\rightarrow$ generation
(created)

ex: two-terminal device



.. look at the polarity of the voltage

.. drops 20V in the direction
of the arrow
↙ drop

$$\therefore P = +VI$$

$$= +20 \cdot 2$$

$$P = 40 \text{ W}$$

.. positive power : element is dissipating
40W

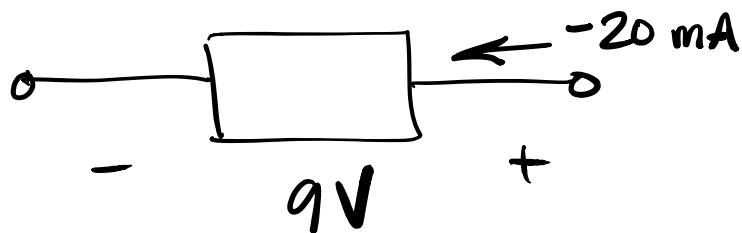
.. absorbs 40W and must do something
with it!

.. resistor : gives off heat

.. motor : produce mechanical work

.. lamp : visible light (+ heat)
etc..

ex:



the key: signs before numbers!

.. voltage drops in direction of current flow

$$\therefore P = +VI$$

$$= +9 \cdot -20 \text{ mA}$$

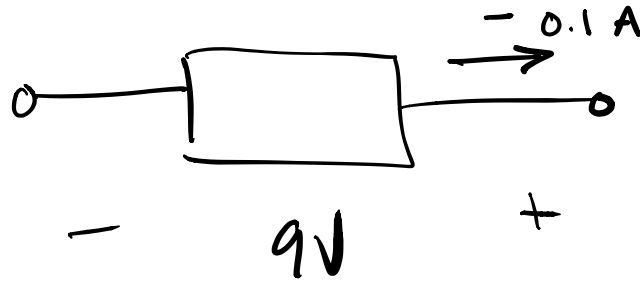
$$P = -180 \text{ mW}$$

.. power is negative; so we could write

$$P = 180 \text{ mW} \text{ generated}$$

.. this is probably a 9V battery,
generating power which is then
dissipated by some other device

ex:



∴ voltage rises 9V in the direction of current

$$\therefore P = -\overset{\text{rise}}{\downarrow} V I$$
$$= -9 \cdot -0.1$$

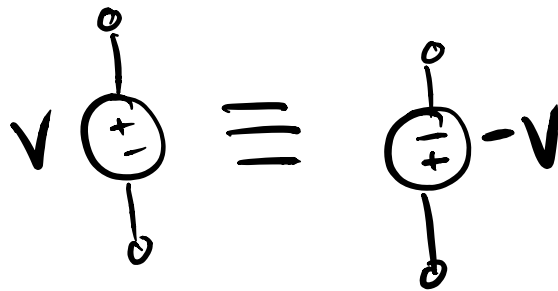
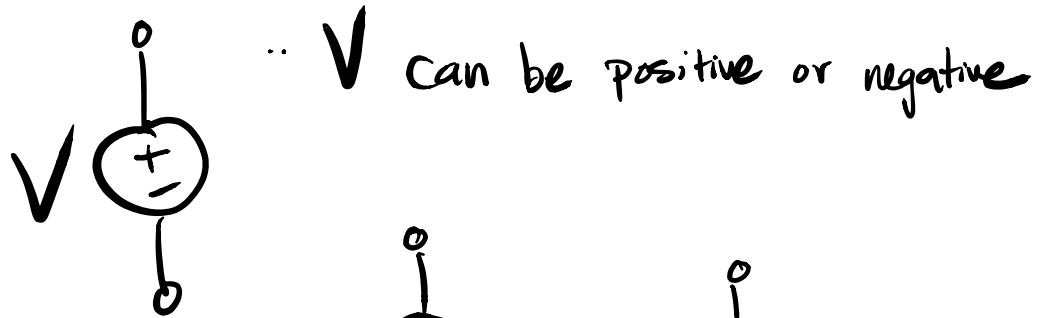
$$P = +0.9 \text{ W (dissipated)}$$

∴ this could be that same 9V battery being charged; meaning, it dissipates that 0.9W by storing energy

∴ do not assume dissipation or generation until you compute the power and look at the sign!

Voltage source

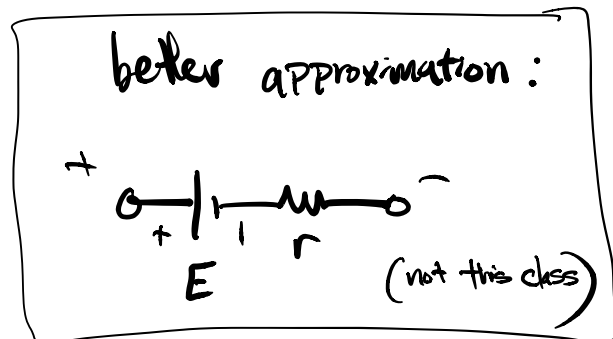
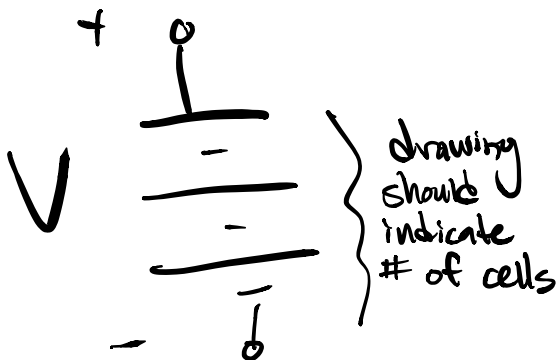
specifically: ideal independent voltage source



.. voltage across voltage source is fixed,
regardless of current through it
(unless invalid circuit; i.e., shorted)

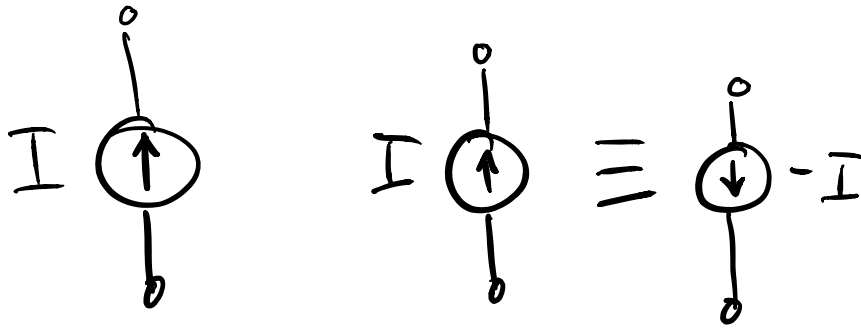
battery: often modelled as ideal independent
voltage source

.. this class: treat it as one!



Current source

.. ideal independent current source



.. current is fixed, regardless of voltage,
if circuit is valid (i.e., no open-circuit)

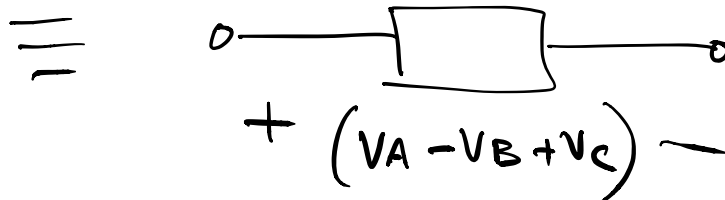
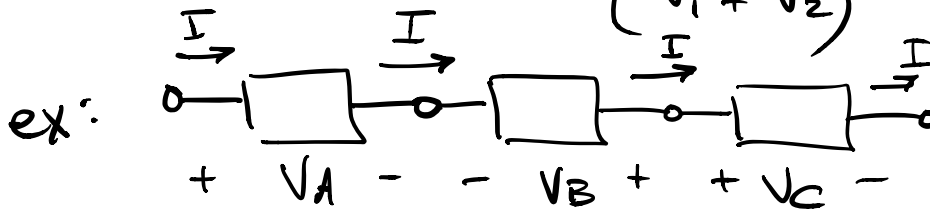
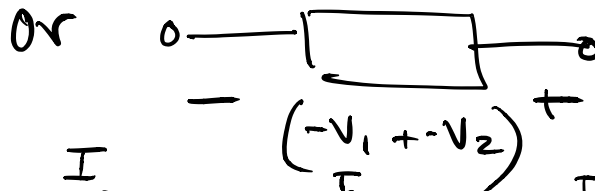
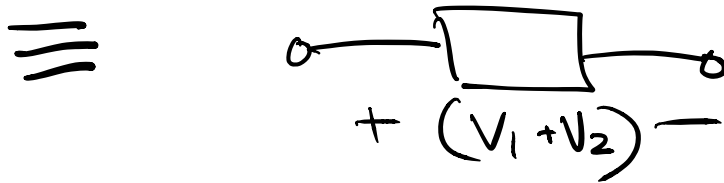
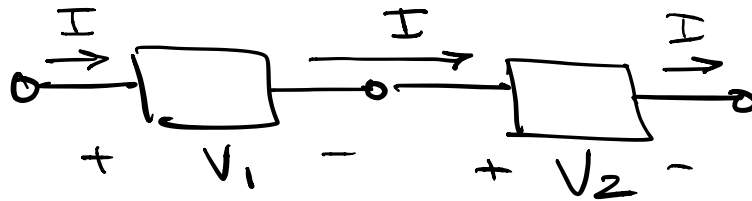
.. any source may either dissipate or generate power : don't assume!!

.. if it dissipates, often called a
"sink" rather than source
(probably not this class)

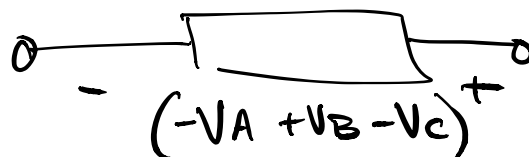
devices in series :



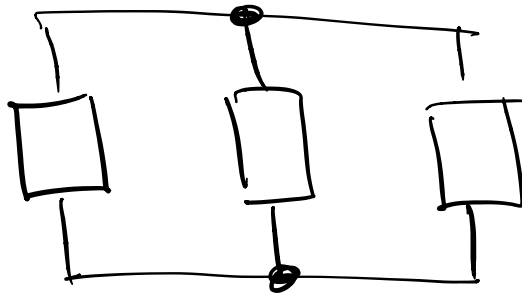
.. voltages add (watch polarities!)
current is the same through each device



OR

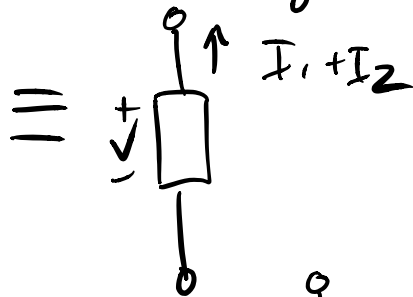
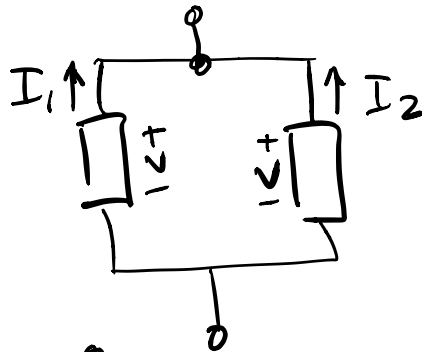


devices in parallel :

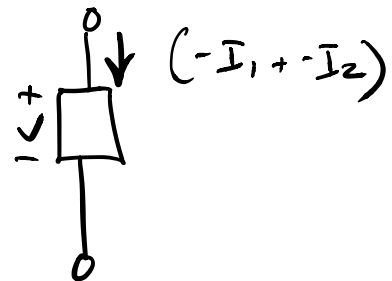


.. currents add
 (watch directions!)
 .. voltage is same
 across each device

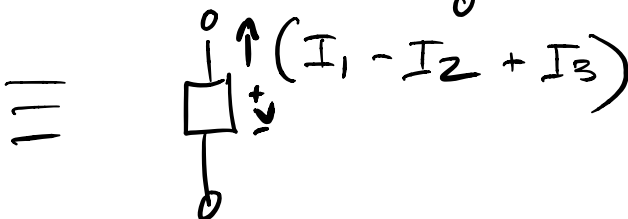
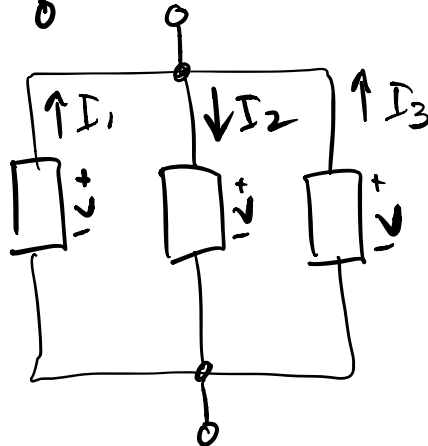
ex:



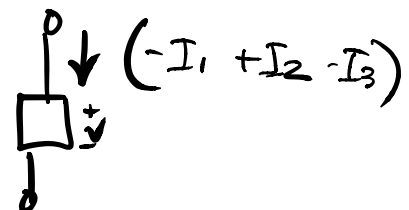
OR



ex:



OR



Homework: Read Ch.1, 2.1-2.3

ex: determine power in each element

